

Thm. Suppose that f is analytic in $\mathbb{C}_\infty$, except for i so, singularities $z_1, \dots, z_n, \infty$.

Then, $\text{Res}(f, \infty) + \sum_{k=1}^n \text{Res}(f, z_k) = 0$.

Proof. By the hypothesis, for some $\delta > 0$, a Laurent's representation

$$f(z) = \sum_{n=-\infty}^{\infty} a_n z^n$$

is valid in $\delta < |z| < \infty$. Specifically, put $\delta = \max\{|z_k| \mid k=1, 2, \dots, n\}$.

Choose $R > \delta$, and γ be a curve, $\{z \in \mathbb{C} \mid |z| = R\}$, the clockwise direction.

And, for $|z| = R$, the Laurent's representation of f converges uniformly, so that

$$\int_{\gamma} f(z) dz = \sum_{n=-\infty}^{\infty} a_n \int_{\gamma} z^n dz = -2\pi i \cdot a_{-1}.$$

so, define $\text{Res}(f, \infty) = -a_{-1}$.

Moreover, $f(z) = \sum_{n=-\infty}^{\infty} a_n z^n$ is analytic for $|z| > R$

iff $f(\frac{1}{z}) = \sum_{n=-\infty}^{\infty} a_n z^{-n}$ is analytic for $0 < |z| < \frac{1}{R}$ ($R < \frac{1}{|z|} \Leftrightarrow |z| < \frac{1}{R}$).

So $a_{-1} = \text{Res}\left(\frac{1}{z^2} \cdot f\left(\frac{1}{z}\right), 0\right)$. $\therefore$ Actually, just definition.

Example. Let $f(z) = 1/(z-3) \cdot (z^n - 1)$ ($n \in \mathbb{N}$). Evaluate $I = \frac{1}{2\pi i} \int_{|z|=2} f(z) dz$.

By the Residue Theorem, $I = \sum_{i=1}^n \text{Res}(f, \exp(2\pi i \cdot \frac{k}{n})) = -\text{Res}(f, \infty) - \text{Res}(f, 3)$

Since $\text{Res}(f, \infty) = -\text{Res}\left(\frac{1}{z^2} \cdot f\left(\frac{1}{z}\right), 0\right) = -\text{Res}\left(\frac{1}{z^2} \cdot \frac{z^{n+1}}{(1-z)(1-z^n)}, 0\right) = 0$,

because the rational function is analytic at $z=0$.

Consequently,

$$I = -\text{Res}(f, 3) = -\lim_{z \rightarrow 3} \frac{1}{z^{n-1}} = \frac{1}{1-3^n},$$

because f has simple pole at $z=3$.